

Semantic Event Detection using Optimized Vision–Language Model

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Task Information

Task: Semantic Event Detection with Optimized Vision–Language Model

Implementation: Standalone Python

Model Used: OpenCLIP (ViT-B/32)

1 Objective

The objective of this project is to develop a Python pipeline that detects semantic events from a local video file using a Vision–Language Model (VLM) and demonstrate model optimization for real-time inference using dynamic quantization.

2 Model Selection

The model used is **OpenCLIP (ViT-B/32 architecture)**, trained on large-scale image-text pairs.

Reasons for Selection:

- Strong zero-shot classification capability
- Lightweight architecture suitable for CPU inference
- Efficient image-text embedding generation
- Compatible with optimization techniques

The model performs:

- Video frames $\rightarrow$ Image embeddings
- Event prompts $\rightarrow$ Text embeddings

Cosine similarity is computed between embeddings to determine the most relevant semantic event.

3 Input and Output

Input

- Local video file: `input_video.mp4`
- Predefined event prompts:
 - “a person walking”
 - “vehicle stopping”
 - “crowded scene”

Output

- Frame-level semantic prediction
- Console-based detection logs
- Performance metrics

4 Processing Pipeline

1. Load OpenCLIP model
2. Extract frames from video
3. Preprocess frames
4. Encode image features
5. Encode text features
6. Compute cosine similarity
7. Select highest similarity event
8. Output detection results
9. Measure inference time

5 Model Optimization

Dynamic quantization was applied using:

```
torch.quantization.quantize_dynamic()
```

Quantization converts model weights from FP32 to INT8 format, resulting in:

- Reduced model size
- Faster inference
- Lower memory usage

6 Performance Comparison

Metric	Original Model	Quantized Model	Improvement
Model Size (MB)	577.18	343.22	40.5% Reduction
Inference Time (sec)	0.00093	0.00050	Faster
Device	CPU	CPU	Same

Table 1: Performance Comparison Before and After Quantization

7 Results

- Total Frames Read: 229
- Frames Processed: 22
- Detected Event: *a person walking*
- Model Size Reduced by 40.5%
- Improved inference latency

8 Trade-offs

- Possible minor precision loss due to INT8 quantization
- CPU-only execution (no GPU acceleration)
- Frame sampling reduces computational load

9 Project Folder Structure

semantic-event-detection/

```
main.py
original_model.pth
quantized_model.pth
input_video.mp4
requirements.txt
README.md
report.pdf
```

10 Submission Checklist

- Working Python script
- Optimized model file
- Performance comparison table
- Technical report (1–2 pages)
- GitHub repository link

11 GitHub README Content

Overview

This project detects semantic events from video using OpenCLIP and demonstrates optimization through dynamic quantization.

Features

- Frame-level semantic detection
- Zero-shot classification
- Dynamic INT8 quantization
- Performance benchmarking

Performance Results

Metric	Original	Quantized
Model Size	577 MB	343 MB
Inference Time	0.00093s	0.00050s

How to Run

```
pip install -r requirements.txt
python main.py
```

12 requirements.txt

```
torch
open_clip_torch
opencv-python
numpy
```

13 Final Submission Checklist

- Code runs without error
- Quantized model included
- Performance table included
- Report PDF included
- GitHub repository created
- Public repository link ready

14 Conclusion

The project successfully demonstrates semantic event detection using a Vision–Language model and shows that dynamic quantization significantly improves efficiency while maintaining functionality.

This solution is suitable for edge devices, CPU-only systems, and real-time applications.